Automated Video Moment Retrieval from Natural Language Questions

COYEN Laurene
Georgia Institute of Technology
lcoyen3@gatech.edu

EL BOUZEKRAOUI Younes
Georgia Institute of Technology
ybouzekraoui3@gatech.edu

BECHT Dunstan
Georgia Institute of Technology
dbecht3@gatech.edu

MORATA Jules
Georgia Institute of Technology
jmorata3@gatech.edu

Abstract

We propose an approach yielding time windows in videos to answer questions based on content analysis. On the one hand, our approach uses images-to-text to predict descriptions of the video sequences. These descriptions are used as inputs of a pre-trained BERT model, along with questions, to predict time windows. On the other hand, video subtitles are input on the same pre-trained BERT model to generate other time windows. These intervals are merged using a random forest regressor and the results are evaluated. The final precision from our work is 32% *on the training set and* 22% *on the test set.*

1. Introduction/Background/Motivation

As there is more and more video content available on the Internet it feels necessary to be able to efficiently search through it. Thus, the objective of this project is to develop a tool to search through videos, using simple questions. Given a video and a question, we want to get the time window (start time and end time) in the video where the question is answered.

At the moment, such a process does not really exist on a public scale. While videos can be found through browsing and keyword searches, these methods rely on video titles and tags rather than the actual content of the videos.
Some research has been conducted to detect objects or actions in videos, in order to generate these descriptive tags. This can be helpful for classical search but, again, the only thing returned would be the video, not the time window of interest in the video. Similarly, some research is led using the sound or meta-text attached to the video like subtitles or even a description.
Our approach introduces a novel method that combines various techniques in order to leverage the full scope of available data and return the time window of interest of the video. In fact, before 2017, many papers such as Antol et al (2015) [1] showed progress in visual question answering given a single image as input. They used a combination of computer vision and natural language processing techniques. From 2017, people have been showing more and more interest in video QA that was yet unexplored. Actually, creating a dataset for question answering on videos is a tough task for annotators that need to consider subtitles, descriptions, questions, answers, time windows... In 2017, Zeng et al [12] introduced a novel model leveraging video descriptions, using LSTMs to encode the temporal information. Yu et al [11] assess the task of multiple-choice question answering which is the most common way of approaching the problem. However, Zhao et al [9] propose an approach generating open-ended answers. They encode the information from the videos and the questions with a bidirectionnal LSTM, and used two attention mechanisms. In fact, it is necessary to pick the relevant frames and accumulate information over time. In 2020, Yan et al [10] used a BERT approach to encode the subtitles and a "sequence of visual concepts". However, studies do not focus on approaching the task to output a time window containing the answer.

There are many examples of applications which can be made of a tool to search through videos such as a student looking to review a particular part of a 2-hour lecture, someone wanting to watch some part of a long conference, or even someone who wishes to look at a certain event in a TV show replay. There are numerous use cases for such an application and having an efficient tool to perform this type of search would allow easier and more complete access to the information, following the trend initiated by the popularization of the Internet and web browsers during the past decades.

For our task, a dataset containing videos with annotations such as subtitles and descriptions, as well as questions and answers on them was needed. We found TVQA, a large-scale dataset based on 6 popular TV shows (Friends, The Big Bang Theory, How I Met Your Mother, House M.D., Grey's Anatomy, Castle). Noting the lack of research on this subject, a team of researchers created this dataset as a benchmark[5]. Its first version was released in 2018. TVQA contains 152.5K Question-Answer pairs from 21.8K video clips, over 460 hours of video divided into training, validation, and testing sets. Each TV show is represented in short clips. In fact, for each TV show, the data is divided in seasons, episodes, segments (before or after the opening song), and clips. The data that we use consists of:

1. the subtitles for each clip, along with the time windows delimiting the dialogues,
2. several descriptions for each clip, describing actions delimited by time window intervals,
3. the video frames (3 per second),
4. a series of questions for each clip along with the correct answer and the time window in the clip delimiting the passage with the correct answer. The answers require a deep understanding of the video content and general knowledge about the characters and the video.

Figure 1. An example of moment localization from natural language query in the TVQA dataset.

2. Approach

Instead of focusing on one characteristic of the video, our approach aims at combining the use of the images and the text of the video. At first, we assume that our method should be applicable to any video as long as subtitles and video frames are provided. Thus, we forbid to use the descriptions provided in the dataset. We combine several methods such that:

1. The first part generates descriptions from images.
2. The second part takes these predicted descriptions as input, understands it with natural language processing (NLP) and outputs a time window.
3. The third part takes the subtitles as input, interprets it with NLP and outputs a time window.
4. The fourth part combines these time windows to output one time window.

For clarity, the pipeline of the model we propose is shown in 2.

Each step will be detailed in the following sections.

2.1. Image processing

First idea: perform action localization to produce a description Even though we first thought of doing object detection through the videos, the first models we worked on were action localization networks as we believed that they would produce better descriptions than object detection approaches.
We first spent a lot of time on trying to use the R-C3D network [8] but we struggled to install all the required environment which is not up to date and so we faced too many problems, mainly setting up the Caffe framework [4].
Thus, we switched to another architecture which is called ActionFormer [13]. Despite encouraging results after training, we realized that this model needed a particular step of feature extraction as a preprocessing and we could not find any explanation on how to perform it correctly on our dataset as the one which was used to train the model, ActivityNet [3], already included the extracted features. Thus we were neither able to perform the inference on our Dataset TVQA nor to fine tune it on this same dataset.

Second idea: extract descriptions from video frames In the light of what was explained above we chose to implement a simpler method which is to give a description of some frames of the video and to use these descriptions to produce one for the whole clip by concatenation. However, running this on every single frame of a video would have been too heavy, both in computation time and output file size. Thus, we chose to take 1 frame every 6 seconds, which is 1 frame out of 18 as we have 3 frames per second. Then we attribute the description of each frame to the 6 seconds window (or maybe less in the case of the last one) centered around the studied frame. We decided to start with the frame at 3s as it limited the need to consider different cases. The following sampling process is described in 3.

To perform this image-to-text task we used a Vision Transformer concatenated with GPT-2 proposed on Huggingface which is a library based on Pytorch that provides users an API to access pretrained models and datasets, the pretrained model is the following.

2.2. Text and Audio processing

The model In this subsection, we explain how we search for an answer in the descriptions generated by the previous approach and the subtitles of the video, which represent the audio. Our task is Question Answering. Hence, we need a model that can extract the meaning of the text and answer questions. We proposed to use the BERT

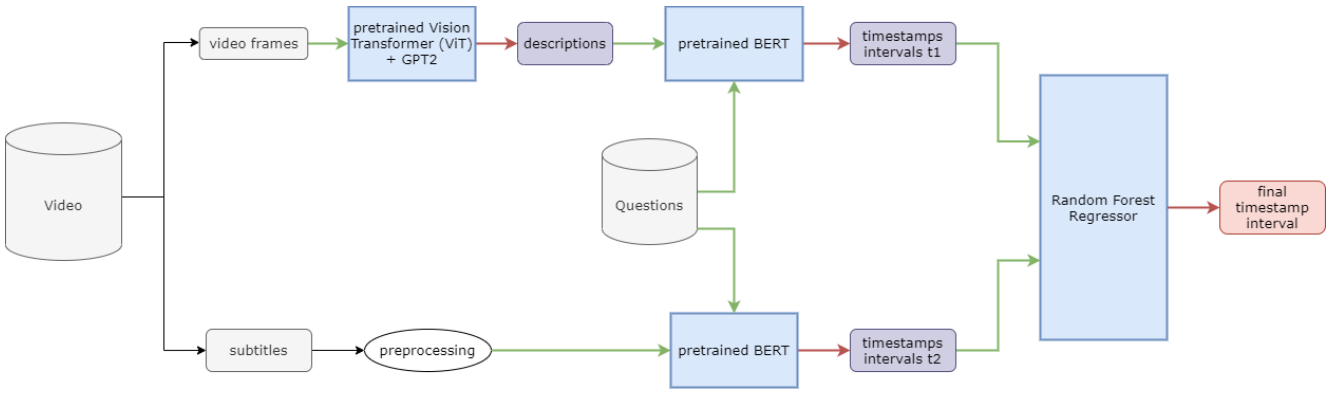

Figure 2. General pipeline of our model. The green and red arrows represent respectively inputs and outputs to or from a model. The data is in gray. The models used are in blue and the intermediary data generation are in purple. The final output is the final time window in red.

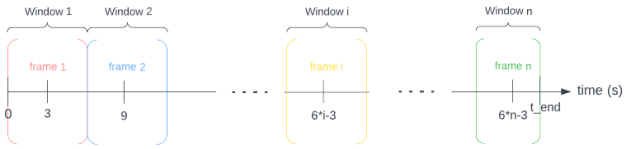

Figure 3. Frames sampling process

model, which stands for Bidirectional Encoder Representations from Transformers. BERT was introduced in 2019 by researchers from Google [2], it uses transformers to understand the context of the words. Unlike previous language models, BERT is bidirectional: relying on the words passed the current one, but also the upcoming ones. BERT became a popular choice since it achieves SOTA results on a wide range of NLP tasks, including question answering.

The Task For our task, we keep in mind that what is interesting to us is the final time window interval corresponding to the passage. We first thought of a model that outputs probabilities for the different time window intervals. That is, if the text considered is the subtitles, the possible intervals would correspond to the different time intervals for the lines in the subtitles. Then we would output the time interval which corresponds to the subtitle that answers best the question. That being said, we realized a similar task was performed with BERT for Question Answering. Specifically, in our approach, we assume the answer is a substring of the passage, there is no need to generate an answer. Therefore, our task is Span-based Question Answering. We found a BERT model pretrained on the SQuAD 2.0 dataset (Stanford Question Answering Dataset) [6]. This dataset gathers 100k question/answer pairs on text from more than 500 Wikipedia articles and 50k questions that cannot be answered. The pretrained model predicts the answer along with start and end indices of the answer in the passage.

For clarity, here is an example:

Context: The game was played on February 7th, 2016, at Levi's Stadium in the San Francisco Bay Area at Santa Clara, California.

Question: What year was the game?

Answer: 2016

In our approach, data is filtered by TV show, season, episode, segment and clips. For each clip, we concatenate all the descriptions. We keep a mapping of the indices of each description segment and the corresponding time windows. In this way, the concatenation constitutes the input context to the model, and the time window will be easily retrieved from the output. The same concatenation is done for the approach with the subtitles. In practice, the pre-trained model from HuggingFace was taken from here.

The anticipated issues The method is based on the assumption that the answer is present in the context. This is true for the descriptions provided with the data (the "gold" descriptions). This remains true for the descriptions generated from the previous method of image processing, depending on the quality of the results. However, it is expected to be more problematic in the subtitles. In fact, originally the subtitles are on the form "Mary: Today is going to be a good day!": the answer will not necessarily be found in the subtitles. In order for the network to interpret better the ":", we chose to do some preprocessing to reformulate the subtitles as "Mary said 'Today is going to be a good day!'". Since BERT is able to understand the deep meaning of a sentence, thanks to its transformers-based architecture, we still expect a positive outcome from this method.

2.3. Results merging

At this step, we have time windows generated from the subtitles and from the generated descriptions. In order to combine them, we propose a simple regression method. We use a Random Forest Regressor that takes in input the predicted intervals and outputs another time window. The

Random Forest is trained on 80% of the time windows data, with the real answer windows as a training objective, and tested on 20%. This approach aims at improving the precision of the interval, reducing the gap between the predicted time window and the goal one. Note that the predicted interval no longer corresponds directly to a particular subtitle or a particular description, but it is generated. In practice, we utilized the scikit-learn library to implement the Random Forest.

Our results are reported in the following section.

3. Experiments and Results

3.1. The metric

Due to the particular nature of the expected result, it seemed appropriate to develop a new metric of accuracy adapted to the problem. We try to quantify here how close a time interval predicted by the model is to the expected time interval. A first idea would have been to reward the model with 1 if the intervals are close and 0 otherwise. But this raises the question of the differentiability of the function. Then we can consider the time of the expected interval covered by the prediction. But such an approach favors models that predict large time intervals. So we tried to opt for a solution that both takes into account the overlap and favors the model that gives a time interval similar to the expected one in terms of duration. This was done by dividing the overlap time of the two-time intervals by the maximum duration of the two time windows. Thus, a model that predicts exactly the expected time window obtains an accuracy of 1. If the model prediction contains the expected time window, but its prediction is spread over too large a duration, then its accuracy is reduced by the too-large duration in the denominator. If the model misses the time window, then the accuracy is 0. All this is reflected in the following precision function:

$$\mathrm{Acc}(a,b) = \frac{\max(0, \max(t_0^a, t_0^b) - \min(t_1^a, t_1^b))}{\max(t_1^a - t_0^a, t_1^b - t_0^b)} \quad (1)$$

This metric is in fact really similar to the well known IoU [7] but in one dimension. We chose our metric instead of the standard IoU because it was easier for us to compute and it has a very similar behavior. Indeed, when either one interval is contained in the other or their intersection is empty, our metric is strictly equal to the IoU. Otherwise, the shapes of the functions are the same except that IoU penalizes more this case than our metric as the union of the intervals is larger than the wider one.

3.2. Image processing and Caption Generation

For the first approach which was to the ActionFormer [13] model which is a Self-attention based Transformer model known to give impressive results on Action detection and localization within a video, we first trained on single-class action in order to reproduce results presented in the paper, then trained on multi-class (200 Types of Action Classes in ActivityNet Dataset) over 100 epochs on Tesla T4 GPU (Using AWS EC2) for about 6 hours and obtained the following Training Loss curve.

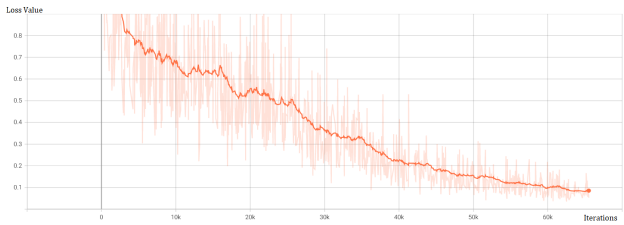

Figure 4. Loss function of ActionFormer model over 100 epochs of the ActivityNet Dataset

We tested the trained model on sample videos of ActivityNet to evaluate the model's performance and post-processed the output to obtain a starting and ending time for each Action that was detected in the video, one of the issues we found in the output is that many occurrences of the same action in overlapping time windows in addition to low confidence ($\leq$ 30%) of many detected actions. The solution we proposed is to eliminate the actions with confidence less than a threshold (25%) and to merge the remaining overlapping time windows.

The main issue we encountered using the previously described approach is that in order to test the model on our Dataset we were in need of features extracted from TVQA videos Dataset, those features were provided only for the ActivityNet dataset, and they are in a specific format that was neither clearly documented nor explained. Thus we were not able to extract those features from our Dataset and thus the inference was not possible using the TVQA Dataset.

For the second approach, we've used a pretrained ViT concatenated with GPT-2. Unfortunately we could not fine-tune it as we planned to do because we did not have labeled data corresponding to the problem (i.e. descriptions associated to frames). However, with more time we could have either labeled some data manually or run some unsupervised training or even perform RL-like training with some human evaluation of the description as a reward. But trying this would have endangered the rest of the project as we were not sure to have the time to test the following parts, so we just ran the model on our data from TVQA and obtained correct results from an human evaluation point of view.

{"vid_name": "castle_s03e05_seg02_clip_10", "duration": 90.66666666666667, "ts": [36.0, 42.0], "desc": ["people in a room"]}
{"vid_name": "castle_s03e05_seg02_clip_10", "duration": 90.66666666666667, "ts": [42.0, 48.0], "desc": ["a man in a suit and tie looking at the camera"]}
{"vid_name": "castle_s03e05_seg02_clip_10", "duration": 90.66666666666667, "ts": [48.0, 54.0], "desc": ["a man with a beard is looking at the camera"]}
{"vid_name": "castle_s03e05_seg02_clip_10", "duration": 90.66666666666667, "ts": [54.0, 60.0], "desc": ["a man standing in a kitchen holding a wine glass"]}
{"vid_name": "castle_s03e05_seg02_clip_10", "duration": 90.66666666666667, "ts": [60.0, 66.0], "desc": ["a man in a black shirt and a white bow tie"]}
{"vid_name": "castle_s03e05_seg02_clip_10", "duration": 90.66666666666667, "ts": [66.0, 72.0], "desc": ["a man and a woman sitting at a table"]}

Figure 5. Example of Time Windows and Description generated using Vision Transformer + GPT2 Model on TVQA Dataset

3.3. Evaluation of approaches

We evaluate each approach individually and in the next subsection, we evaluate the combination of the approaches.

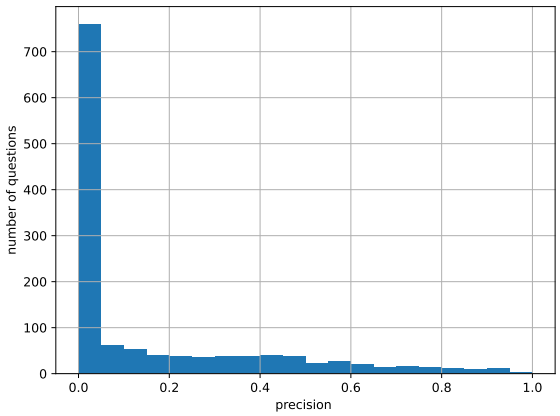

Figure 6. Model based on exact descriptions

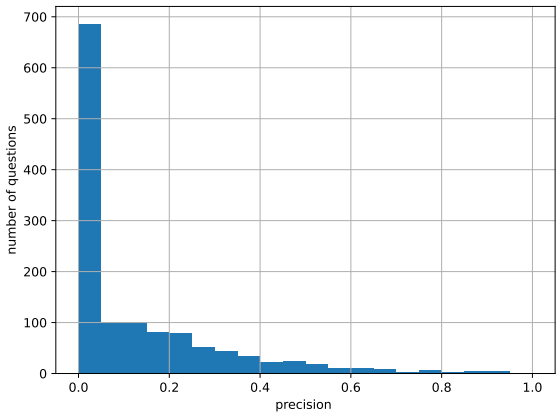

Figure 7. Model based on subtitles

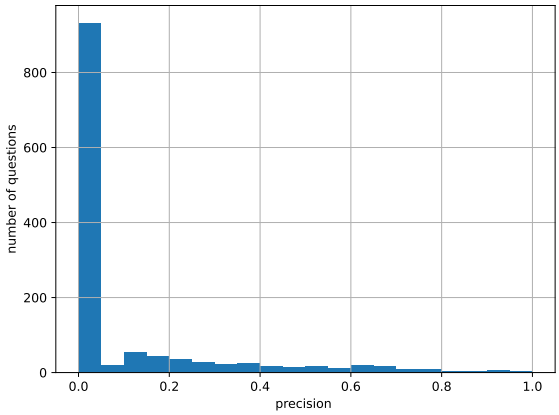

Figure 8. Model based on predicted descriptions

Figures 6 to 8 show the number of responses by precision range. The displayed accuracy corresponds to the one introduced in equation (1). We can observe that a majority of predicted time windows have no, or very few, intersection with the expected one. On average, the model based on exact descriptions has a precision of 15%, the one based on subtitles has a precision of 12 % and the one based on predicted descriptions has a precision of 10 %.

3.4. Evaluation of time windows combination

As explained in subsection 2.3, we trained a Random Forest Regressor to combine the time windows. We evaluate our method using 5-fold cross validation with the metric specified in subsection 3.1 which, for simplicity, is called "Intersection over Union" in Plot 9. Comparing to the results shown in subsection 3.3, the combination of time windows improved the precision of each method individually: while the average precisions of the methods based on the subtitles and the predicted descriptions were respectively

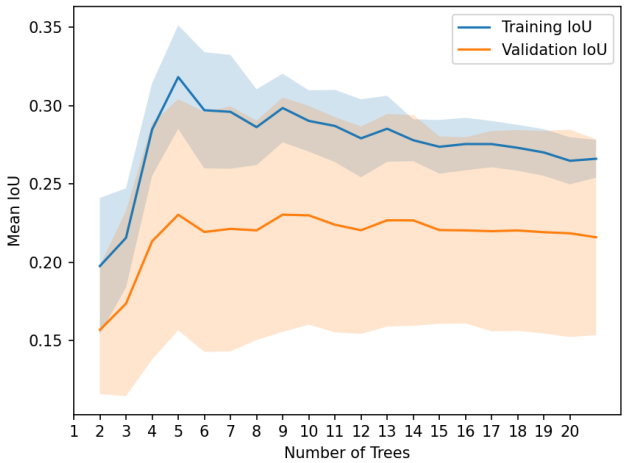

Figure 9. Precision over the number of trees used in the random forest, on 5-fold cross-validation

12% and 10%, the combination with regression gives a 32% precision on the training set and a 22% on the test set. So it seems that the RF Regression compensates some bias of our model.

At the training level, grid search was performed to find the best hyper-parameters: no max-depth, 4 as the minimum threshold of samples needed for a leaf node, 10 as the minimum number of samples required to split an internal node. According to Figure 9, 5 trees seem to be providing the best precision. Then, the precision is slightly shrinking for both the training and testing set, this may indicate a slight overfitting. Overall, the precision is quite poor combining the different approaches. As this model is only performing regression, this is certainly due to the time windows predictions. However, a positive outcome is that the combination of these methods yielded better results than each method used individually. Therefore, the method of combining time windows proved success while the time windows prediction need to be improved.

4. Conclusion

The main objective of this work was to experiment on a novel architecture to leverage the full scope of videos for temporal localization on question answering. We evaluated our approaches with a novel metric, similar to Intersection over Union, and found that the combination of time windows allowed to get a better result than the individual methods. In this regard, we feel that we succeeded. However, we did not provide as good final IoU measures from the approaches as we expected. Overall, despite a few setbacks, we succeeded in follow through with our approach.

4.1. Observations and limitations

The evaluations are directly performed on the output time windows. But it would be wise to evaluate the prediction of the image processing model at first. Improving these predictions would allow an improvement of the time windows predictions. However, as stated by Zeng et al [12], while generating relevant, non-specific, descriptions is fairly easy, it is not that easy to evaluate their quality since they are open-ended.

4.2. Further investigation

As a way of improving our results, we implemented a training of the pretrained BERT model on our generated description in order to finetune the model on our data and obtain better results. However, given the long training time, we only trained our model on a few episodes, thus no significant improvement was made. Finetuning our model on the subtitles and generated descriptions could help the model to better understand our data. In particular, the format of the subtitles is specific, and the results could improve if the model was trained on it. However, it is true that, given that the answer is not necessarily written in the subtitles, the model might not perform well. One idea would be to try a different model for question answering on the subtitles, that is able to deduce and generate an answer but still retrieving the time interval.

4.3. For practical use

We produced a first pipeline for query search through video, which still needs to be improved in term of accuracy but is operational. Now if we want to deploy it on a public scale to act like a web browser, we would need to :

1. Generate the subtitles with Speech-To-Text if they don't exist

2. Loop through a dataset of videos instead of just searching in one video

References

[1] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C Lawrence Zitnick, and Devi Parikh. Vqa: Visual question answering. In *Proceedings of the IEEE international conference on computer vision*, pages 2425–2433, 2015. 1

[2] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*, 2018. 3

[3] Fabian Caba Heilbron, Victor Escorcia, Bernard Ghanem, and Juan Carlos Niebles. Activitynet: A large-scale video benchmark for human activity understanding. In *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 961–970, 2015. 2

[4] Yangqing Jia, Evan Shelhamer, Jeff Donahue, Sergey Karayev, Jonathan Long, Ross Girshick, Sergio Guadarrama, and Trevor Darrell. Caffe: Convolutional architecture for fast feature embedding. *arXiv preprint arXiv:1408.5093*, 2014. 2

[5] Jie Lei, Licheng Yu, Mohit Bansal, and Tamara L Berg. Tvqa: Localized, compositional video question answering. *arXiv preprint arXiv:1809.01696*, 2018. 2

[6] Pranav Rajpurkar, Robin Jia, and Percy Liang. Know what you don't know: Unanswerable questions for squad. *CoRR*, abs/1806.03822, 2018. 3

[7] Hamid Rezatofighi, Nathan Tsoi, JunYoung Gwak, Amir Sadeghian, Ian Reid, and Silvio Savarese. Generalized intersection over union: A metric and a loss for bounding box regression, 2019. 4

[8] Huijuan Xu, Abir Das, and Kate Saenko. R-C3D: region convolutional 3d network for temporal activity detection. *CoRR*, abs/1703.07814, 2017. 2

[9] Hongyang Xue, Zhou Zhao, and Deng Cai. Unifying the video and question attentions for open-ended video question answering. *IEEE Transactions on Image Processing*, 26(12):5656–5666, 2017. 1

[10] Zekun Yang, Noa Garcia, Chenhui Chu, Mayu Otani, Yuta Nakashima, and Haruo Takemura. Bert representations for video question answering. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, March 2020. 1

[11] Youngjae Yu, Jongseok Kim, and Gunhee Kim. A joint sequence fusion model for video question answering and retrieval. In *Proceedings of the European Conference on Computer Vision (ECCV)*, September 2018. 1

[12] Kuo-Hao Zeng, Tseng-Hung Chen, Ching-Yao Chuang, Yuan-Hong Liao, Juan Carlos Niebles, and Min Sun. Leveraging video descriptions to learn video question answering. *Proceedings of the AAAI Conference on Artificial Intelligence*, Feb. 2017. 1, 6

[13] Chenlin Zhang, Jianxin Wu, and Yin Li. Actionformer: Localizing moments of actions with transformers, 2022. 2, 4